Family Connect: Product and Technical Specification

Oct 14, 2026 · @Rana · Working name; v0.1 for the development team

1. Overview and goals

Family Connect is a voice-first AI companion that keeps older adults socially engaged at home and strengthens, rather than replaces, their contact with adult children who live away or are busy.

The problem. Loneliness affects roughly 1 in 5 older adults in India and about a third of adults aged 50–80 in the US, and is linked to higher risk of dementia, heart disease, stroke and depression. Many older people cannot or will not use smartphone apps, and their children cannot be present every day.

The product. The parent talks by voice to an AI avatar on a tablet or TV, and to Alexa+ on Echo devices throughout the home. The avatar is either a clearly labelled digital version of their son or daughter (created and approved by the child) or a neutral companion with its own name. It chats, remembers their life, plays real messages from family, reminds them about medicines and tests, detects emergencies and escalates to real people. Alexa+ extends the experience to every room: help phrases, messages, reminders, everyday tasks and smart-home safety. The adult child receives daily reports, conversation starters and timely nudges to call or visit.

Goals for v1

- Parent can start a conversation with zero touch, by saying a wake phrase.
- Daily engaging conversation in Hindi, English or a mix.
- Real calls and visits from family increase after onboarding (primary success metric).
- Emergencies reach a real person within minutes, with layered backup.
- Parent consent and privacy are enforced in code, not just policy.

Non-goals for v1

- Avatars of deceased people (excluded by design).
- A general-purpose companion for young people.
- Medical advice, diagnosis or dose changes.
- Building custom hardware.

2. Product principles

These are design constraints. Any feature that conflicts with them needs explicit sign-off.

1. **Support connection, never replace it.** The avatar fills gaps between real contact and actively pushes toward calls and visits. Success is measured by real human contact, not avatar usage time.
2. **Always honest that it is AI.** The child-avatar is labelled on screen at all times, refers to itself as the "digital" version, and setup includes a family conversation explaining it.
3. **Defer, don't decide.** The avatar never makes commitments, decisions or promises on behalf of the real child. Plans, money, health and family matters are passed to the real person.
4. **The child controls their avatar.** Only the child can create their avatar, from their own photo and voice, with explicit consent. They set boundaries and can read what "they" said.
5. **The parent controls their data.** The parent agrees to what is shared in reports and can keep things private.
6. **Safety never depends on the AI alone.** Emergency detection runs as a deterministic engine alongside the LLM, with layered escalation to humans.
7. **Remind, never advise on health.** The system relays doctors' instructions and reminders; it never interprets symptoms, changes doses or recommends treatment.
8. **Alexa+ as a core channel, with no single point of failure.** Alexa+ on Echo devices is a standard part of every installation and the parent's voice access in every room. Safety-critical paths are always duplicated through the companion app and our own telephony, so no outage or platform change on one channel can stop an emergency response.

3. Users and roles

Role	Who	Interface	Key permissions
Parent	Older adult living at home, possibly with hearing, vision or early cognitive issues	Voice only: companion app on tablet/TV, and Alexa+ on Echo devices in other rooms	Talk to avatar; ask for help; approve what is shared; pause or stop the service

Role	Who	Interface	Key permissions
Primary child	Adult son or daughter, account owner and payer	Family app (Android/iOS)	Create own avatar; set boundaries; view reports and logs; set reminder intervals; approve orders; manage emergency contacts
Family circle member	Siblings, grandchildren, relatives	Family app	Receive reports if parent consents; call; send messages; optionally create own avatar
Backup contact	Neighbour or local relative	Phone (SIP call), optional app	Receives emergency calls when family cannot be reached
Response partner	Private ambulance or medical response service	Phone / API	Receives emergency dispatch with address and details
Admin / support	Family Connect staff	Internal console	Device support, escalation audits; no access to conversation content without consent

4. Feature requirements

Priority: P0 = pilot must-have, P1 = Phase 2, P2 = later.

4.1 Avatar and companion

ID	Requirement	Priority
F-01	Two modes: child-avatar (created by the child from their own photo, optional voice clone with consent) and neutral companion (own name, generic look). Family chooses per parent; can use both.	P0

ID	Requirement	Priority
F-02	Persistent on-screen label on child-avatar, e.g. "Digital Priya". Avatar periodically refers to itself as the digital version.	P0
F-03	Child-set persona brief: how they talk, nicknames, topics to avoid, things they would never say.	P0
F-04	Defer, don't decide: on plans, money, health, family disputes or promises, avatar replies "Let me check with the real me" and creates a deferral item for the child.	P0
F-05	Conversation log and summary visible to the child for everything said in their name.	P0
F-06	Dementia safeguard: family (ideally with doctor) can switch child-avatar off and default to neutral companion.	P0
F-07	Hand-over: when the real child is available or calling, avatar steps aside ("The real Priya is calling!").	P0

4.2 Conversation and engagement

ID	Requirement	Priority
F-10	Natural voice conversation in Hindi, English and code-mixed speech; regional languages later.	P0
F-11	Long-term memory of the parent: people, places, life history, interests, routines, recent events.	P0
F-12	Engagement activities: reminiscence prompts, daily news and cricket chat, festivals, recipes, old songs and films, quizzes, word games, antakshari, prayers on request.	P0
F-13	Life-story capture: stories recorded (with consent) into a family memory book.	P1
F-14	Outward nudges: suggest telling family a story, local groups, temple or community events.	P1

ID	Requirement	Priority
F-15	Honest behaviour: can disagree gently, admit not knowing, never invents facts about the real child.	P0

4.3 Family connection

ID	Requirement	Priority
F-20	Child records voice/video messages; avatar plays them and captures spoken replies.	P0
F-21	App-to-app video calls (WebRTC) between family app and tablet; parent answers by voice.	P0
F-22	Daily report to child: mood, activity, topics discussed, deferral items, conversation starters, one-tap "Call Mum".	P0
F-23	Smart call reminders: child-set interval plus earlier nudges when parent seemed low, unwell or lonely.	P0
F-24	Good-time detection: suggest calls when parent is awake and free and child is in their availability window (time-zone aware).	P1
F-25	Family circle and rota: spread contact across siblings and grandchildren.	P1
F-26	Visit tracking: days since last visit, occasions (birthdays, festivals, appointments), visit countdown for parent.	P1
F-27	Shared activities over video: meal together, watch match, cook a recipe, prayer, games.	P2
F-28	Family phone number: relatives without the app call a number that rings the tablet (SIP inbound).	P1

4.4 Safety and emergency

ID	Requirement	Priority
F-30	Voice help phrases in every room: "Hello Priya, help" on the tablet; "Alexa, tell Priya I need help" on any Echo. Both route to the same safety engine.	P0
F-31	Deterministic detection: distress keywords, no response to check-ins, unusual silence, missed routines; camera-based fall detection on device.	P0 (fall: P1)
F-32	Escalation for concerns that are not emergencies: primary child → other family → backup contact, by call and WhatsApp.	P0
F-33	Emergency camera view: with prior consent, live view shared only with the responder during an active escalation; visible and spoken indicator; no recording by default.	P1
F-34	Whole-home coverage via Echo devices with Alexa+ in bedroom and bathroom areas (standard install).	P0
F-35	Fall prevention with Alexa smart-home routines: night lights switch on when the parent gets up at night (motion sensor or voice).	P1
F-36	Automatic emergency dialling: on high-confidence triggers only (parent asks for an ambulance or emergency help; help request or fall followed by no response), spoken 10-second countdown with "say cancel", then calls 112 or 108 automatically.	P0
F-37	Parallel calling in emergencies: emergency number, primary child and backup contact are called at the same time, not in sequence.	P0
F-38	Live person on the line: emergency call is bridged into the tablet speaker and microphone so the operator can talk to the parent; child conferenced in when they answer; short recorded opener with address only.	P0
F-39	Emergency details pushed to child, backup and response partner by call and WhatsApp: address, key medical information, medicines, trigger.	P0

4.5 Health reminders and medicines

ID	Requirement	Priority
F-40	Child scans prescription or report in family app; AI extracts medicines, doses, timings, test dates; child must confirm before use.	P0
F-41	Medicine and test reminders via avatar, Echo notifications, and child reports.	P0
F-42	Voice reorder: parent asks; system reorders exactly what the prescription states via partner pharmacy; child approves before payment; avatar confirms delivery.	P1
F-43	No medical advice: questions about symptoms, skipping or changing medicines go to child or doctor.	P0

4.6 Consent and privacy controls

ID	Requirement	Priority
F-50	Parent consent capture at setup, by voice and witnessed by family, for each data category.	P0
F-51	Parent can mark topics private ("Don't tell Priya about this"), except safety concerns.	P0
F-52	Child consent for avatar likeness and voice, revocable at any time.	P0
F-53	Pause / sleep mode by voice; listening indicator always visible.	P0

4.7 Alexa+ experience

ID	Requirement	Priority
F-60	Alexa skill with account linking set up from the family app; Echo devices registered to the household.	P0

ID	Requirement	Priority
F-61	"Alexa, talk to Priya" starts an avatar session on the tablet; on audio-only Echo it starts a voice conversation with the same persona, memory and boundaries.	P0
F-62	"Alexa, any messages from Priya?" plays family voice messages and records a spoken reply.	P0
F-63	Proactive notifications on Echo: new messages, upcoming calls, medicine and test reminders, delivery updates.	P0
F-64	Medical appointments and tests confirmed by the child sync to a calendar linked to the parent's Alexa, so Alexa+ can remind and answer "When is my blood test?" natively.	P1
F-65	Everyday life through Alexa+: music and bhajans, news, cricket, smart home, groceries (Amazon Now), food and travel services. No custom build; covered in onboarding.	P0
F-66	Alexa-to-Alexa calling and Drop In as a backup family-calling channel, with Drop In permissions approved by the parent.	P1
F-67	Native Alexa+ agent via the Multi-Agent SDK, so the companion persona runs inside Alexa+ conversations without a skill launch phrase.	P2

4.8 Wellbeing updates and WhatsApp delivery

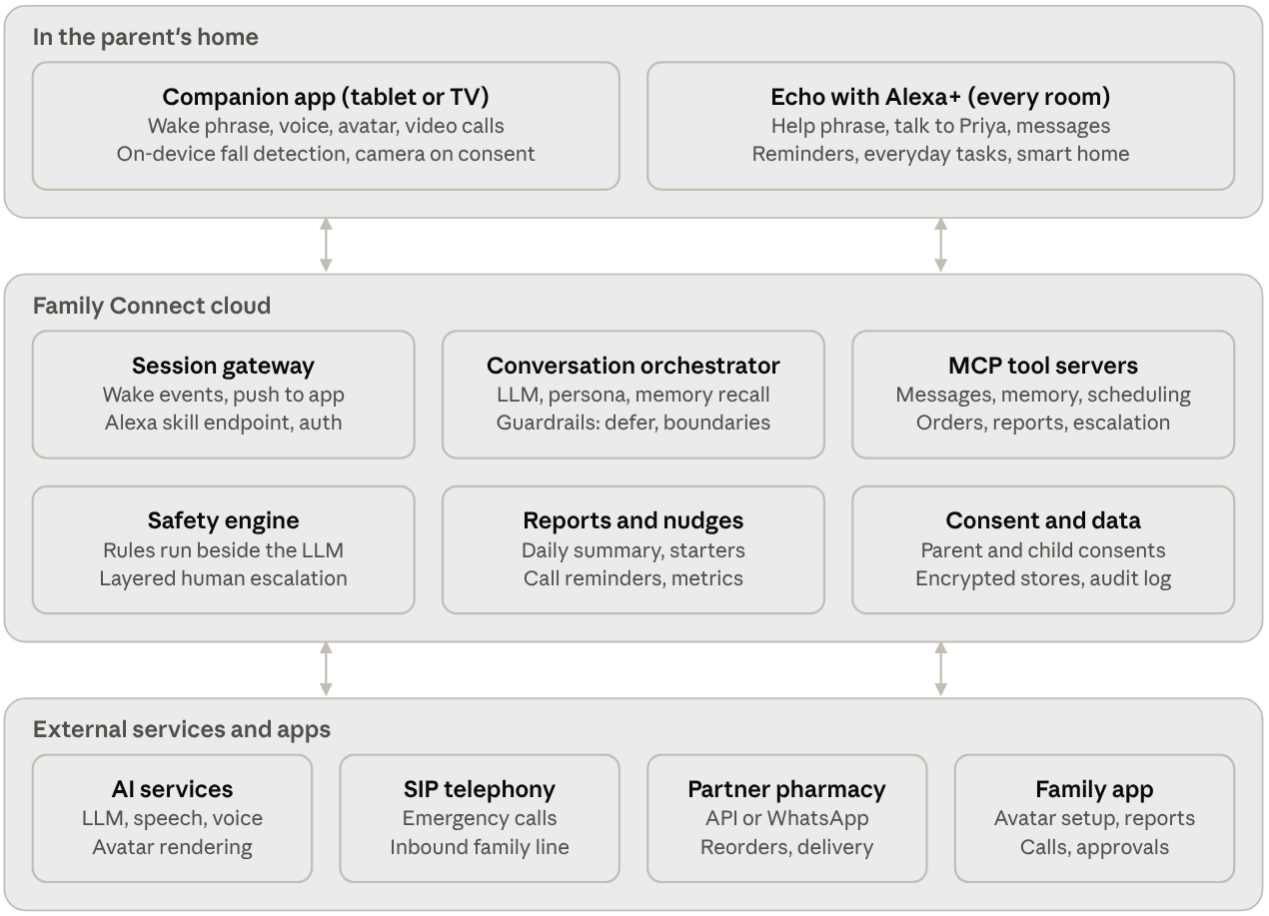
ID	Requirement	Priority
F-70	Scheduled wellbeing digest at a time set by the child (daily or weekly): mood, energy, sleep and appetite mentions, activity, topics, medicine adherence, open deferrals, 2–3 conversation starters, "Call Mum" button.	P0

ID	Requirement	Priority
F-71	AI-triggered alerts in three tiers: **Urgent** (immediately, overrides quiet hours: distress, confusion, emergency, parent asking for the child); **Important** (within the hour, respects quiet hours: missed medicines two days running, illness signs, unusually low mood, not eating); **Worth knowing** (added to the next digest).	P0
F-72	Each update available as a short text summary and a 30–60 second voice note; child chooses text, voice or both.	P0
F-73	Delivery by WhatsApp through the official WhatsApp Business Platform, plus in-app and push. Child opts in during setup.	P0
F-74	WhatsApp messages use pre-approved templates, carry the voice note and quick-reply buttons ("Call Mum", "Reply with voice message", "I'll call tonight"), and link to the full report in the family app.	P0
F-75	Sensitive health detail stays in the app; WhatsApp carries only a short summary. Parent privacy flags apply; safety items are always included.	P0
F-76	Alert fatigue controls: per-child quiet hours, tier thresholds and frequency caps; urgent alerts always go through.	P1

5. System architecture

The product runs on two connected front ends in the home: the companion app (avatar, camera, calls) and Alexa+ on Echo devices (whole-home voice, notifications, everyday tasks, smart-home routines). Both talk to the same cloud backend and MCP tools, so memory, boundaries and safety rules are identical on both. Safety-critical paths never depend on a single channel.

Two front ends in the home share one cloud brain



system architecture · 3 layers, 12 components

5.1 Components

Component	Responsibilities	Suggested technology
Companion app	Kiosk-mode app on tablet or TV; on-device wake phrase; streaming speech in/out; avatar rendering; WebRTC calls; on-device fall detection; listening and camera indicators; receives push to start sessions	Android (Kotlin); on-device wake-word engine; on-device pose model for falls; FCM push
Echo devices with Alexa+ (standard)	Help phrase and "talk to Priya" from other rooms; reminders and message notifications; everyday Alexa tasks	Alexa custom skill now; Alexa+ Multi-Agent SDK when available in India

Component	Responsibilities	Suggested technology
Session gateway	Authenticates devices; receives Alexa skill requests and app wake events; starts and routes sessions; pushes wake signals to the app	API gateway + WebSocket service
Conversation orchestrator	Builds context (persona, boundaries, memory, today's events); runs the LLM; pre- and post-response guardrail checks; calls MCP tools	LLM provider behind an internal interface (swappable); guardrail classifier
MCP tool servers	All actions the AI can take, with permission and consent checks enforced inside each server (section 7)	MCP servers per domain
Safety engine	Consumes event stream (speech flags, silence, routines, fall events, help phrases); runs rules independent of the LLM; drives escalation state machine	Event bus + rules engine + state machine
Reports and nudges	Daily summaries, conversation starters, call reminders, good-time detection, connection metrics	Scheduled jobs + LLM summarisation
Consent and data	Stores consents and privacy flags; enforces them on every read; audit log; encryption	Relational DB, vector store for memory, object storage for media; India-region hosting
AI services	Speech-to-text and text-to-speech (Hindi, English, code-mixed); voice clone (consented); avatar generation and lip-sync	Third-party vendors behind adapters
SIP telephony	Outbound emergency calls with IVR; bridge to tablet; inbound family number	Licensed Indian SIP trunk provider
Partner pharmacy	Prescription reorders and delivery status	Partner API or WhatsApp Business ordering

Component	Responsibilities	Suggested technology
Family app	Avatar creation and consent; boundaries; reports and logs; messages; calls; approvals; prescription scanning	Cross-platform mobile (Flutter or React Native)

5.2 Hardware for the pilot

- **Primary:** Android tablet (LTE model preferred for backup connectivity) on a stand, plus a USB or Bluetooth conference speakerphone for far-field hearing.
- **Standard:** Echo Dot with Alexa+ in the bedroom and near the bathroom; optional Echo Show in the kitchen or second living area. Linked to the parent's Amazon account during setup.
- **Later:** big-screen option on Android TV or Fire TV Cube with USB webcam. Avoid Vega OS Fire TV Sticks (4K Select and later), which cannot run Android apps.

Audio rule: each conversation uses one device for both speaking and listening. At the tablet, the avatar's video and voice both come from the tablet with a conference speakerphone (loud speaker plus echo-cancelling microphones). In other rooms, the parent talks to Priya by voice on the Echo, and Alexa can offer to move the conversation to the tablet. Do not split video on the tablet and voice on an Echo: Bluetooth delay breaks lip-sync, a microphone on one device cannot cancel a speaker on another, and Alexa skills cannot stream real-time conversation in sync with another device.

6. Key flows

6.1 Starting a conversation

1. Parent says "Hello Priya" to the tablet (on-device wake phrase), or "Alexa, talk to Priya" to an Echo.
2. For Echo: Alexa skill request → session gateway → push to tablet app.
3. App wakes screen, shows the labelled avatar (or neutral companion), opens a streaming voice session.
4. Orchestrator loads persona, boundaries, memory and today's context (messages waiting, reminders due, recent mood).
5. Conversation runs; safety signals stream to the safety engine in parallel.
6. Session ends on silence or "bye"; orchestrator writes memory updates and report notes.

6.2 Deferral ("Will you come for Diwali?")

1. Guardrail classifier tags the turn as a commitment request.
2. Avatar replies with a deferral ("Let me check with the real me and get back to you").
3. `create_deferral` tool posts the item to the child's app with context.
4. Child answers in the app (text, voice note or call).
5. Avatar relays the real answer next session, attributed to the real child.

6.3 Emergency escalation

1. Trigger: help phrase (tablet or Echo), distress detected, fall detected, or no response to a check-in after a set window.
2. Safety engine opens an incident and checks with the parent by voice ("Are you okay?").
3. **Not an emergency** (parent responds and declines help, or concern is low): follow-up check and Important alert to the child by call and WhatsApp (F-32).
4. **Emergency** (parent asks for an ambulance or help; or help request or fall followed by no response): spoken countdown, "Calling emergency services in 10 seconds. Say 'cancel' if you're okay."
5. At zero, in parallel: dial 112 or 108 (cellular route on LTE tablet preferred, SIP if verified); SIP call to primary child with IVR; SIP call to backup contact; dispatch to response partner where configured.
6. Emergency call is bridged into the tablet speaker and microphone so the operator talks to the parent; recorded opener states the address; child is conferenced in when they answer.
7. With prior consent, camera view becomes available to the child or responder; indicator on screen and spoken notice.
8. Address, medical summary, medicines and trigger are sent to child, backup and response partner by WhatsApp and in-app.
9. Every step is logged; incident closes only when a human confirms the parent is safe.

6.4 Medicine reorder

1. Parent: "Hello Priya, my BP tablets are running out."
2. Orchestrator calls `check_prescription` and `get_refill_status`.
3. `draft_order` creates an order for exactly the prescribed item and quantity with the partner pharmacy.
4. Child receives an approval request; nothing is paid until approved.

5. On approval, order is placed; avatar confirms to the parent; Echo notification on delivery day.

6.5 Prescription or report scan

1. Child photographs the document in the family app.
2. Extraction service returns medicines, doses, timings and test dates with confidence scores.
3. Child reviews and edits every field; low-confidence fields are highlighted.
4. On confirmation, reminders are scheduled and the prescription is stored for reorders.

6.6 Daily report and nudge

1. Throughout the day, the orchestrator tags wellbeing signals from conversations (mood, energy, sleep, appetite, pain or illness mentions, worries) and adherence events.
2. Alert engine scores each signal against the three tiers (F-71); Urgent and Important alerts are sent immediately as text and voice note via WhatsApp, push and in-app.
3. At the child's chosen time, the digest job summarises the period: mood, activity, topics, deferrals, adherence, and 2–3 conversation starters from the parent's own stories.
4. Text-to-speech generates a 30–60 second voice note of the digest.
5. Delivery through a pre-approved WhatsApp template with voice note, quick-reply buttons and a link to the full report in the app.
6. Call nudge: checks call cadence against the child's interval and today's mood; adds a nudge if due or if the parent seemed low.
7. Parent privacy flags are applied before sending; safety items are always included; sensitive health detail stays behind the app link.

7. MCP tool design and data model

All AI actions go through MCP tool servers. The same tools serve the companion app, the Alexa skill/agent and any future channel, and the LLM provider can be swapped without changing them. **Permission, consent and boundary checks are enforced inside each server, never left to the model.**

7.1 Tool servers

Server	Tools (initial)	Enforced inside the server
family-messages	`list_messages`, `play_message`, `record_reply`, `send_message`	Sender/recipient in family circle

Server	Tools (initial)	Enforced inside the server
memory	`recall`, `remember`, `get_profile`	Parent privacy flags; no memory of private-marked topics in reports
boundaries	`check_topic`, `get_persona`	Child's avoid-list and persona; blocks disallowed topics
deferrals	`create_deferral`, `get_answers`	Always routes to the real child; avatar cannot resolve its own deferrals
scheduling	`propose_call`, `get_availability`, `start_call`	Child availability windows; time zones
health	`get_reminders`, `log_adherence`, `check_prescription`, `get_refill_status`	Only child-confirmed data; read-only for the LLM except adherence logging
orders	`draft_order`, `get_order_status`	Exact prescribed item and quantity only; payment only after child approval
reports	`add_report_note`, `add_starter`	Privacy flags; safety notes always allowed
safety	`raise_concern`, `start_escalation`, `get_incident`	LLM can raise or start; only the safety engine can close; rules also trigger independently

7.2 Core data entities

Entity	Key fields
Household	id, address, language preferences, devices, timezone
Parent	profile, health summary, consents, privacy flags, routines, emergency info
FamilyMember	role, contact numbers, availability windows, report preferences, reminder interval
Avatar	owner (child), mode (child / neutral), likeness assets, voice consent, persona brief, avoid-list, status

Entity	Key fields
Consent	subject, category, granted_at, witnessed_by, revoked_at
Session	channel (app / Echo), start/end, transcript ref, summary, mood estimate, safety flags
MemoryItem	type (person, place, story, preference, event), content, source session, privacy flag, embedding
Deferral	question, context, status, child answer, relayed_at
Message	from, to, media ref, played_at, reply ref
Prescription	document ref, medicines (name, dose, timing), tests, confirmed_by, confirmed_at
Reminder	type, schedule, channel, adherence log
Order	items, pharmacy, approval status, approver, delivery status
Incident	trigger, timeline of escalation steps, responders, camera access log, resolution, closed_by
ContactEvent	type (call, video, visit, message), participants, duration — feeds success metrics

8. Integrations

8.1 Alexa+ and Echo (core channel)

Alexa+ is installed in every household and gives the parent voice access in every room. It carries four roles; the avatar and the safety decisions stay in our backend, reached through the same MCP tools.

Role	What it does	How it connects	Status / constraint
Whole-home safety	Help phrase from any room; routes to the safety engine	Custom skill → session gateway → safety engine	Feasible now with classic skill

Role	What it does	How it connects	Status / constraint
Second front door	"Talk to Priya", family messages, replies	Custom skill now; Alexa+ Action SDK and Multi-Agent SDK later, all calling our MCP tools	Multi-Agent SDK in preview; apply early
Reminders and notifications	Messages waiting, calls, medicines, tests, deliveries	Skill notifications; calendar linked to the parent's Alexa account	Verify notification availability under Alexa+ in India
Everyday life assistant	Music, news, smart home, groceries, local services, night-light routines	Native Alexa+; we configure routines during onboarding	Alexa+ in India via Early Access; free with Prime, paid otherwise

Integration details

- **Account linking:** OAuth from the skill to Family Connect, completed by the child in the family app during setup.
- **Same brain on both channels:** Alexa requests call the same orchestrator and MCP tools as the app, so memory, persona, boundaries and defer-don't-decide behave identically.
- **Hand-off to screen:** when the parent wants to see the avatar or a video call, Alexa says "I've opened Priya on your tablet" and the gateway pushes the session to the app.
- **Echo Show:** used for Alexa visuals (message cards, reminders) where helpful; the live avatar stays on the companion app.

Known constraints to design around: Echo wake words are a fixed list (no custom names, so the Echo phrase is "Alexa, talk to Priya"); third-party skills cannot render a custom live avatar on Echo Show or access its camera; Echo cannot call regular phone numbers or emergency numbers in India (Alexa-to-Alexa calling only). Amazon discontinued Alexa Together in May 2025, so every safety path through Alexa is duplicated through the companion app and our SIP telephony.

8.2 Calling

Call type	Channel
Family ↔ parent, everyday video	WebRTC between family app and tablet
Concern alert to family or backup	SIP trunk → phone, with IVR; plus WhatsApp

Call type	Channel
Emergency number (112 / 108)	Automatic dial after countdown; cellular call from LTE tablet preferred (correct local routing), SIP only if the provider confirms emergency support and address-based routing
Child and backup in an emergency	SIP calls placed in parallel with the emergency call; conferenced into the tablet
Relative without app calls parent	Inbound family number (SIP DID) → tablet
Medical response partner	Phone or API dispatch with address and medical summary
Backup from other rooms	Alexa-to-Alexa calling via Echo

Use a licensed Indian telecom operator or established cloud telephony provider for the SIP trunk. Households should save the outbound number with a clear name (e.g. "Mum – Family Connect").

Emergency dialling on Android: apps normally cannot place emergency calls without a person confirming in the dialer. On the dedicated tablet, the companion app should hold the default phone app role so it can place the call automatically; developers must verify this on target devices. Run regular emergency drills during the pilot.

8.3 AI services

- **LLM:** behind an internal interface so providers can be swapped; must support tool calling (MCP) and Hindi/English code-mixing.
- **Speech:** streaming speech-to-text and text-to-speech tuned for older voices, Indian accents and code-mixed speech; evaluate on real pilot recordings.
- **Voice clone:** only with the child's recorded consent; watermark generated audio where supported.
- **Avatar:** real-time talking-head rendering from the child's approved photo; target lip-sync latency compatible with natural turn-taking.
- **Document extraction:** OCR + LLM for prescriptions and reports, returning confidence per field.

8.4 Pharmacy

Partner pharmacy via API or WhatsApp Business ordering. Prescription medicines require a valid prescription and pharmacist validation under Indian rules; the system only reorders what a child-confirmed prescription states. AI voice calls to local chemists are out of

scope for v1.

8.5 WhatsApp

- **Official WhatsApp Business Platform** only (Cloud API or an approved provider); never personal numbers or unofficial tools.
- **Opt-in:** each family member opts in during setup in the family app.
- **Templates:** digests, alerts and emergency notices use pre-approved message templates, since businesses cannot send free-form messages outside the 24-hour customer service window.
- **Rich content:** voice-note attachment, quick-reply buttons, and a deep link to the full report.
- **Inbound:** child's voice or text replies are captured and delivered to the parent as family messages (F-20).
- **Fallbacks:** if WhatsApp delivery fails, send push notification and SMS; urgent alerts are also delivered by phone call.

9. Safety, privacy and compliance

9.1 AI safety guardrails

- System design, not just prompts: a guardrail classifier checks each user turn and each avatar response for commitments, health advice, disallowed topics and distress.
- Hard blocks: no medical advice or dose changes; no financial actions; no impersonation outside the labelled avatar; no claims about the real child's life that are not in confirmed data.
- Anti-dependency: avatar regularly bridges outward (family, local groups); reports flag rising avatar time alongside falling real contact.
- Evaluation set: scripted conversations covering deferrals, boundary breaches, confusion, distress and medical questions, run before every model or prompt change.

9.2 Privacy

- On-device wake phrase; audio leaves the device only after the wake phrase or an active session.
- Camera: fall detection runs on device; no video leaves the home except a live view during an active incident, with prior consent and a visible and spoken indicator. No recording by default.
- Parent privacy flags respected in all reports; safety-related information is always shared.

- Child sees logs only of what their avatar said, not the neutral companion's private chats, unless the parent consents.
- Data minimisation, encryption at rest and in transit, India-region hosting, retention limits, and a full audit log of every access to transcripts or camera.

9.3 Consent

- Parent: consent per category (conversation storage, reports, camera in emergencies, health data), captured by voice with a family witness and revocable by voice.
- Child: explicit consent for likeness and voice cloning, revocable at any time (avatar disabled immediately on revocation).
- Capacity: for parents with moderate or advanced dementia, family (ideally with their doctor) decides on child-avatar mode; neutral companion is the default fallback.

9.4 Compliance to review with counsel

- India's Digital Personal Data Protection (DPDP) Act: consent, purpose limitation, data principal rights, and handling of health and biometric (face, voice) data.
- Telecom rules for SIP trunking and connecting internet calls to the public phone network.
- Emergency calling: rules and practice for automated calls to 112 and 108, location routing, and how to avoid false calls.
- WhatsApp Business Platform policies, including what health-related content may be sent in templates.
- Online pharmacy and prescription rules for medicine orders.
- Whether any feature (fall detection, health reminders, wellbeing alerts) could be classed as a medical device; keep claims to "reminders" and "alerts", not diagnosis.
- Call recording disclosure, if incidents are recorded.

10. Non-functional requirements and success metrics

10.1 Non-functional requirements

Targets below are proposed starting points for the team to validate in the pilot.

Area	Requirement
Voice latency	Response begins within about 1.5 seconds of the parent finishing speaking, so turn-taking feels natural

Area	Requirement
Wake reliability	Wake phrase works across the room with TV noise; false wakes tracked and kept rare
Emergency path	Escalation path does not depend on the LLM; first call placed within seconds of a confirmed trigger; continues if the app or LLM is down
Availability	Safety engine and telephony designed for high availability; conversation features may degrade gracefully
Offline behaviour	App announces when it is offline; help phrase still attempts escalation via LTE/SMS fallback where available
Accessibility	Large text and captions, slow and clear speech option, adjustable volume, high-contrast UI
Language	Hindi, English and code-mixed at launch; architecture ready for regional languages
Observability	Per-session metrics, safety-incident tracing, model and prompt versioning
Security	Device attestation, encrypted storage, role-based access, audited staff access

10.2 Success metrics

Metric	Why it matters
Real calls per week and minutes per call (family ↔ parent)	Primary metric: proof that the product increases human contact
Visits per month	Real-world connection
Parent-reported loneliness, measured with a short validated loneliness scale at onboarding and periodically	Outcome measure
Mood trend after real contact vs avatar-only days	Shows where value comes from
Avatar time vs real contact ratio	Guardrail against dependency

Metric	Why it matters
Deferrals answered by children, and time to answer	Health of the defer-don't-decide loop
Medicine and test adherence	Health support value
Incident response time and false-alarm rate	Safety performance
Family retention after 3 months	Business viability

11. Build phases

Durations are left for the team to estimate.

Phase	Scope	Exit criteria
0. Foundations	Consent and data service; MCP server skeletons; LLM and speech vendor evaluation on real older-adult Hindi/English recordings; avatar vendor proof of concept; SIP provider onboarding; emergency-dialling feasibility on target tablets; WhatsApp Business Platform setup and template approval; Alexa skill and account-linking design; apply for Alexa+ Multi-Agent SDK early access; legal review (section 9.4)	Vendors chosen; consent model approved by counsel; latency within target in a lab test; emergency call routing verified

Phase	Scope	Exit criteria
1. Pilot (10–20 families)	Companion app with wake phrase, child-avatar and neutral companion, memory, engagement activities; Echo with Alexa+ as standard install: help phrase, "talk to Priya", family messages, notifications; family app with avatar setup, boundaries, messages, call reminders, prescription scan and reminders; wellbeing digests and tiered alerts by WhatsApp (text and voice); safety engine with automatic emergency dialling and parallel family calls; WebRTC calls	Families use it most days; zero missed escalations in drills from both tablet and Echo; no unintended emergency calls; real calls per week measured against baseline
2. Whole home and health	Alexa calendar sync for appointments; smart-home night-light routines; Drop In and Alexa calling as backup; camera fall detection and emergency view; medicine reorder with partner pharmacy; family circle and rota; good-time detection; visit tracking; inbound family number; alert fatigue controls	Positive change in real-contact metrics; adherence improvement; incident response time within target
3. Scale	Native Alexa+ agent (Multi-Agent SDK); big-screen TV app; regional languages; memory book; shared activities; care-home edition	Retention and unit economics support wider launch

Apply for Alexa AI Multi-Agent SDK early access during Phase 0 so access is in place by Phase 3.

12. Open questions and sources

12.1 To verify before or during Phase 0

- Does the Alexa AI Multi-Agent SDK support the India locale and Hindi, and what visual and notification options does it allow on Echo Show?
- Are Alexa skill notifications available to Indian users under Alexa+ Early Access?

- Can the chosen SIP provider route calls to 112 and 108, and are they routed by the household's address?
- On target Android tablets, can the companion app (as default phone app) place emergency calls automatically over cellular?
- Which private ambulance or medical response partners can accept API or phone dispatch in pilot cities?
- WhatsApp Business Platform provider choice and template approval for wellbeing and emergency messages.
- Partner pharmacy options and their ordering APIs.
- Avatar vendor: real-time latency, Hindi lip-sync quality, data handling and consent terms for likeness.
- Product name and trademark ("Family Connect" is a working name).
- Pricing model: family subscription, hardware bundle, or both.

12.2 Sources

- TechCrunch – Amazon launches Alexa+ in India with Hindi support
- Amazon Developer Blog – Introducing AI-native SDKs for Alexa+
- About Amazon – Alexa Together (discontinued May 2025)
- Asurion – Alexa wake word options
- Android Central – Alexa-to-Alexa calling and Drop In in India
- Amazon help – Sharing files with Alexa+
- Liliputing – Vega OS on Fire TV Sticks
- About Amazon – Amazon Pharmacy India
- WHO – Social connection report (June 2025)
- IAFOR – Loneliness among older adults in India (LASI)
- Neuroscience News – Loneliness and dementia meta-analysis